

Thursday Warm-up: Ampère's Law

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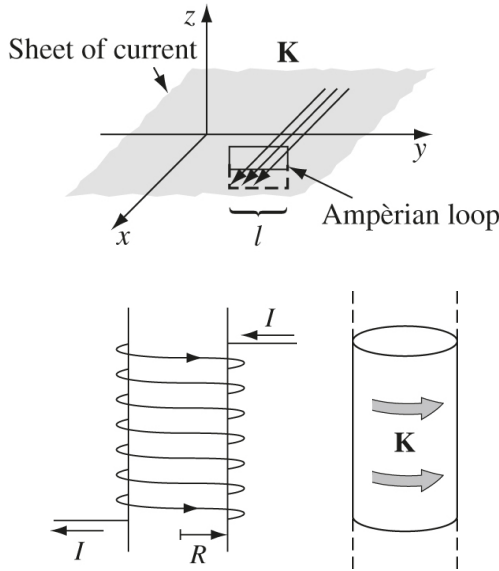


Figure 1: (Top) A sheet of current, $\mathbf{K} = K\hat{\mathbf{x}}$. (Bottom left) A solenoid with finite windings. (Bottom right) A solenoid with high density of windings.

small current in the $+x$ -direction below the sheet?

- Consider Fig. 1 (bottom left and right). (a) Find the $\mathbf{B}$ -field of a very long solenoid, consisting of n closely wound turns per unit length on a cylinder of radius R , each carrying a steady current I . *Hint: use the Amperian loops in Fig. 2 to establish the field inside and outside of the solenoid.* (b) Suppose we shoot a beam of positively-charged particles through the solenoid. Sketch the trajectory of the particles.

1 Memory Bank

- The Biot-Savart Law**, for steady currents:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \int \frac{d\mathbf{l}' \times \hat{\mathbf{r}}}{r^2} \quad (1)$$

- Ampère's Law**, differential form:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad (2)$$

- Ampère's Law**, integral form:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}} \quad (3)$$

2 Ampère's Law

- Consider Fig. 1 (top). Find the $\mathbf{B}$ -field the surface current $\mathbf{K} = K\hat{\mathbf{x}}$, flowing in the xy -plane. (a) First, identify the *direction* of the field, above and below the sheet. Second, calculate the *magnitude* of the field. (b) In which direction would a small current in the $+x$ -direction above the sheet move? What about a

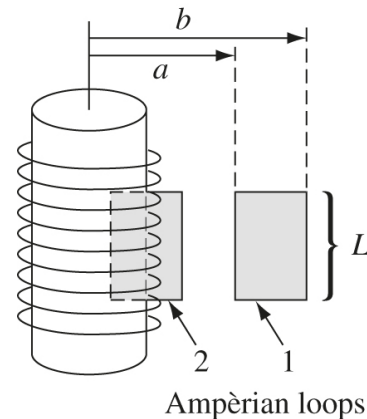


Figure 2: The Amperian loops used for solenoid calculations.